Applied AI Design Workshop

Classic machine learning foundations for faculty-centered AI design

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Workshop overview • Schedule • Learning activities

Day 3 → Day 4 → Day 5

Problem framing, data preparation, model development, evaluation, interpretation, and error analysis.

Workshop Outcomes

NC STATE Engineering

What you should leave prepared to do

Frame applied AI problems

Translate disciplinary questions into prediction tasks with clear inputs, targets, units of observation, baselines, and success criteria.

Prepare useful data

Recognize data-quality issues, create meaningful features, prevent leakage, and structure data for model development.

Select and evaluate models

Choose metrics, compare models against baselines, interpret outputs, and diagnose model errors.

Interpret results

Use residuals, confusion matrices, feature importance, and error patterns to connect model behavior to disciplinary meaning.

Apply in teaching & research

Identify feasible classroom activities and research workflows for classical ML and applied AI adoption.

Core path: problem formulation → data/features → model development → validation → interpretation → improvement

Consolidated Workshop Schedule

NC STATE Engineering

Three-day consolidation of classic ML topics and invited presentations

Day 3	**Foundations**	Problem formulation, regression, classification, and baseline modeling	DK Xu — 30 min
Day 4	**Data and Features**	Data preparation, time-series features, clustering, and PCA	Andrew Petersen — 30 min
Day 5	**Evaluation and Interpretation**	Validation, tree-based models, feature selection, model comparison, and error analysis	Vijay Shah — 30 min

Design principle: each day builds toward a faculty-ready applied AI workflow.

Learning Activities

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How you will engage with the material

Mini-lectures

Concise conceptual framing with applied examples.

Scenario sorting

Match disciplinary problems to ML task types.

Dataset walkthroughs

Inspect, clean, split, and model small tabular datasets.

Metric interpretation

Use residuals and confusion matrices to reason about model quality.

Case discussion

Compare models using accuracy, error patterns, interpretability, and usefulness.

Brainstorming

Capture questions, disciplinary translation issues, and follow-up topics.

Activities emphasize “design judgment” rather than just running a tool.

Day 3 — Foundations of Applied Machine Learning

NC STATE Engineering

Problem formulation, regression, classification, and baseline modeling

Session	Topic / Activity	Speaker
Session 1	From Engineering or Disciplinary Problem to Machine Learning Problem	
Session 2	Classic Machine Learning Overview	
Invited	From Prompts to Agents: Scalable, Efficient, and Grounded Agentic AI for Engineering Workflows	DK Xu
Break	Break	
Session 3	Regression Fundamentals	
Session 4	Classification Fundamentals	
Session 5	Hands-On Activity: Building a Baseline ML Model	

Day 3 Learning Outcomes

NC STATE Engineering

By the end of Day 3, you should be able to:

Translate a disciplinary problem into a suitable ML task.
Distinguish regression, classification, clustering, and forecasting problems.
Establish an appropriate baseline model.
Select initial metrics for evaluating model performance.
Explain how classic machine learning differs from generative and agentic AI.

Day 3 anchors the workshop: clear problem framing makes every downstream AI decision more defensible.

Day 4 — Data Preparation, Feature Engineering, and Model Development

NC STATE Engineering

Data preparation, time-series features, clustering, and PCA

Session	Topic / Activity	Speaker
Session 1	Data Preparation and Feature Engineering for Applied AI	
Session 2	Time-Series Feature Engineering	
Invited	LLM and AI Workflows on NCSU’s Hazel HPC	Andrew Petersen
Break	Break	
Special Topic	Faculty-Selected Applied AI Topic	Facilitated
Session 3	Unsupervised Learning for Discovery	
Session 4	Dimensionality Reduction and Visualization	
Brainstorming	Capture impressions, unresolved questions, and topics requiring discussion	Facilitated

Day 4 Learning Outcomes

NC STATE Engineering

You should be able to:

Identify common data-quality problems before model development.
Prevent data leakage during preprocessing.
Create meaningful features from engineering, scientific, and time-series data.
Apply clustering to unlabeled datasets.
Use PCA to visualize and simplify high-dimensional data.
Identify computing needs for larger AI and LLM workflows.

Day 4 emphasizes the data work that determines whether models learn meaningful patterns.

Day 5 — Model Evaluation, Interpretation, and Selection

NC STATE Engineering

Validation, tree-based models, feature selection, model comparison, and error analysis

Session	Topic / Activity	Speaker
Session 1	Model Evaluation and Experimental Design	
Session 2	Tree-Based Models and Feature Importance	
Invited	NeuralSmith: An Automated AI/ML Engineering Co-Pilot for Domain Scientists and Engineers	Vijay Shah
Session 3	Feature Selection and Model Simplification	
Session 4	Applied Case Study: Selecting and Evaluating an ML Model	
Session 5	Error Analysis and Model Improvement Activity	

Day 5 Learning Outcomes

You should be able to:

Design an appropriate validation strategy.
Select metrics based on the type of prediction problem.
Interpret tree-based models and feature-importance results.
Reduce model complexity through feature selection.
Compare models based on performance, interpretability, and disciplinary usefulness.
Diagnose underfitting, overfitting, systematic errors, and poorly represented cases.

Day 5 turns model outputs into defensible decisions about model quality and usefulness.

Workshop Flow

NC STATE Engineering

A practical sequence for applied AI design

1

Problem

Define objective, target, unit, baseline

2

Data

Prepare, clean, encode, and inspect

3

Features

Create domain-informed and time-series features

4

Models

Train regression, classification, clustering, and tree-based models

5

Evaluate

Use metrics, validation, and error analysis

6

Interpret

Connect model behavior to disciplinary usefulness

You will leave with a repeatable structure for evaluating whether an AI idea is well-framed, data-ready, and model-ready.

Workshop Outputs

NC STATE Engineering

What participants can take back to their courses, labs, and programs

Problem framing template

A structured way to define inputs, outputs, targets, baselines, and success criteria.

Dataset activity pattern

Small reusable exercises for inspection, cleaning, feature creation, modeling, and interpretation.

Evaluation checklist

Questions for metrics, splitting, leakage, baselines, and model comparison.

Teaching examples

Cross-disciplinary examples for regression, classification, clustering, PCA, and error analysis.

Applied AI roadmap

A shared language for faculty AI adoption across teaching, research, and engineering workflows.

Next step: adapt one session activity to a course module, research workflow, or faculty development exercise.